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MIL-M-28787/521
20 SEPTEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-188C/RS232C ASYNCHRONOUS SERIAL CONTROL MODULE, KEY CODE AEK

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-188C/RS232C asynchronous serial control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.538 failures per million operating hours for class I or 9.489 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AEK.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A6	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	-0.12	1.21	-0.12	1.21	-0.12	1.21	mA
D2-D4	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D5	Ibd	V	S	-0.17	1.27	-0.17	1.27	-0.17	1.27	mA
D6	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	-0.12	1.11	-0.12	1.11	-0.12	1.11	mA
D9-D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16,D17	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D18	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D19-D23	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D24	Ibd	V	S	-0.12	0.72	-0.12	0.72	-0.12	0.72	mA
D25	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D26	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D27	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E3	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E4	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E5-E7	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E8-E10	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E11-E13	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E14	Ios	VI	S	-260.0	-15.0	-260.0	-15.0	-260.0	-15.0	mA
E15	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E16	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E17-E21	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E22	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E23	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E24-E27	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E28	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E29	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E30	Ios	VI	S	-7.84	-5.18	-7.84	-5.18	-7.84	-5.18	mA
E31,E32	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E33	Ios	VI	S	-85.0	-15.0	-85.0	-15.0	-85.0	-15.0	mA
E34-E40	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E41	Ios	VI	S	-85.0	-15.0	-85.0	-15.0	-85.0	-15.0	mA
E42-E57	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-5.88	-2.84	-5.88	-2.84	-5.88	-2.84	mA
F2-F4	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F5	Iih	VII	X	-15.2	-4.11	-15.2	-4.11	-15.2	-4.11	mA
F6	Iih	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F7	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F8	Iih	VII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
F9-F13	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F14	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16,F17	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F18	Iih	VII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
F19-F23	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F24	Iih	VII	X	-6.83	-2.75	-6.83	-2.75	-6.83	-2.75	mA
F25	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F26	Iih	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F27-F42	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F43	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G1	Iil	VIII	X	-5.88	-2.84	-5.88	-2.84	-5.88	-2.84	mA
G2-G4	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G5	Iil	VIII	X	-15.2	-4.11	-15.2	-4.11	-15.2	-4.11	mA
G6	Iil	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
G7	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G8	Iil	VIII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
G9-G13	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G14	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G15	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16,G17	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G18	Iil	VIII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
G19-G23	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G24	Iil	VIII	X	-6.83	-2.75	-6.83	-2.75	-6.83	-2.75	mA
G25	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G26	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G27-G42	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G43	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
H1-H56	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J58	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K4	Iozh	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K5	Iozh	XI	X	-1.0	1.0	-1.0	1.0	-1.0	1.0	mA
K6-K9	Iozh	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K10-K17	Iozl	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
L1-L4	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5-L9	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L10	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L14	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L15-L22	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M	PSFC	XIII	S	5.4	10.0	5.4	10.0	5.4	10.0	uF
N	Icc	XIV	X	500	1500	500	1500	500	1500	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1	Ci/o	XVIII	Q	6	175	6	175	6	175	pF
R2	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R3,R4	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R5	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R6	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R7	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R8	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R9	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R10-R14	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R15	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R16-R18	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R19	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R20-R24	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R25	Ci/o	XVIII	Q	8	200	8	200	8	200	pF
R26-R29	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R30	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R31-R34	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R35-R37	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R38-R50	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R51-R66	Ci/o	XVIII	Q	3	75	3	75	3	75	pF
T	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/521-1 for class I and M28787/521-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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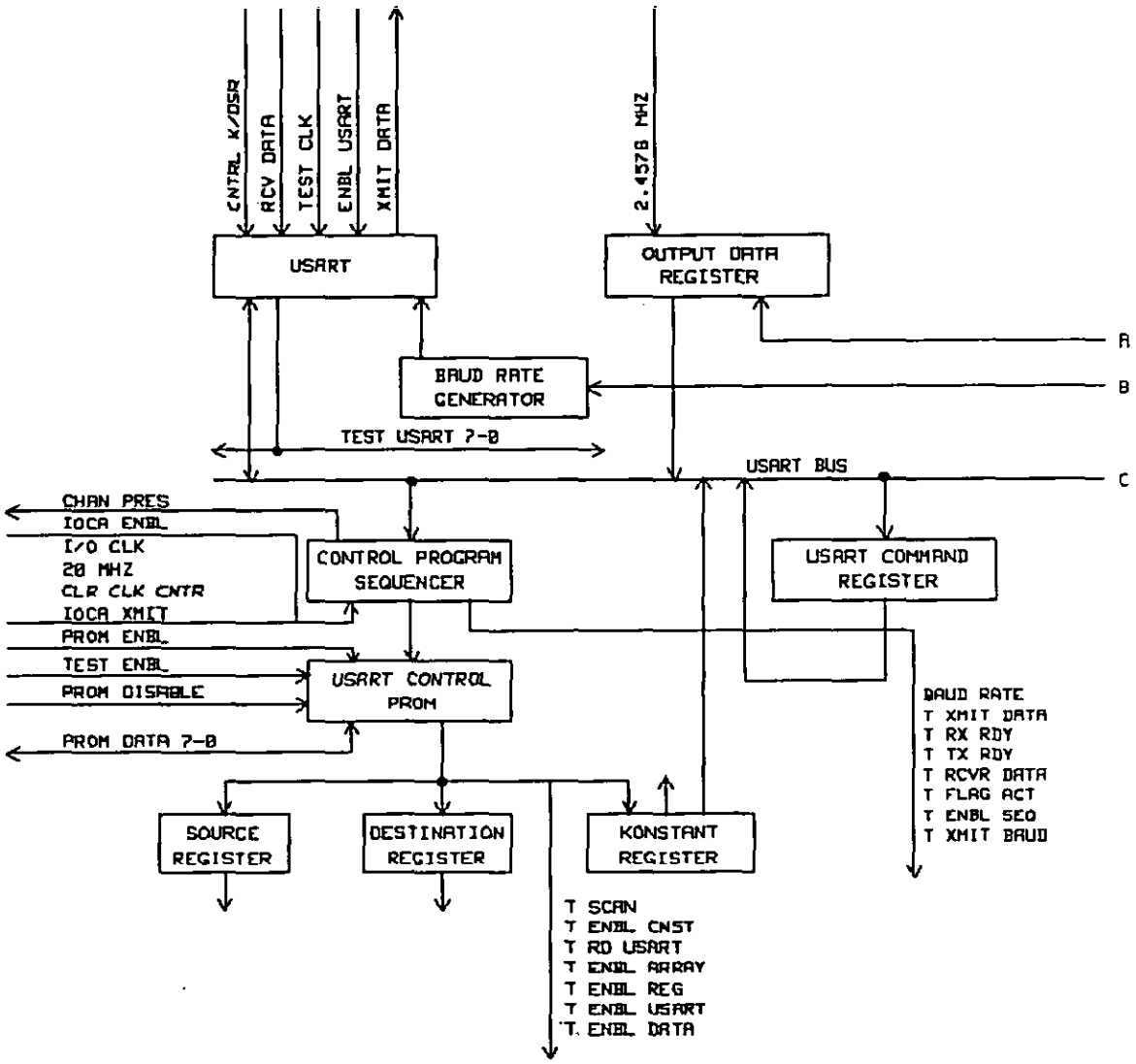


FIGURE 1. AEK functional block diagram.

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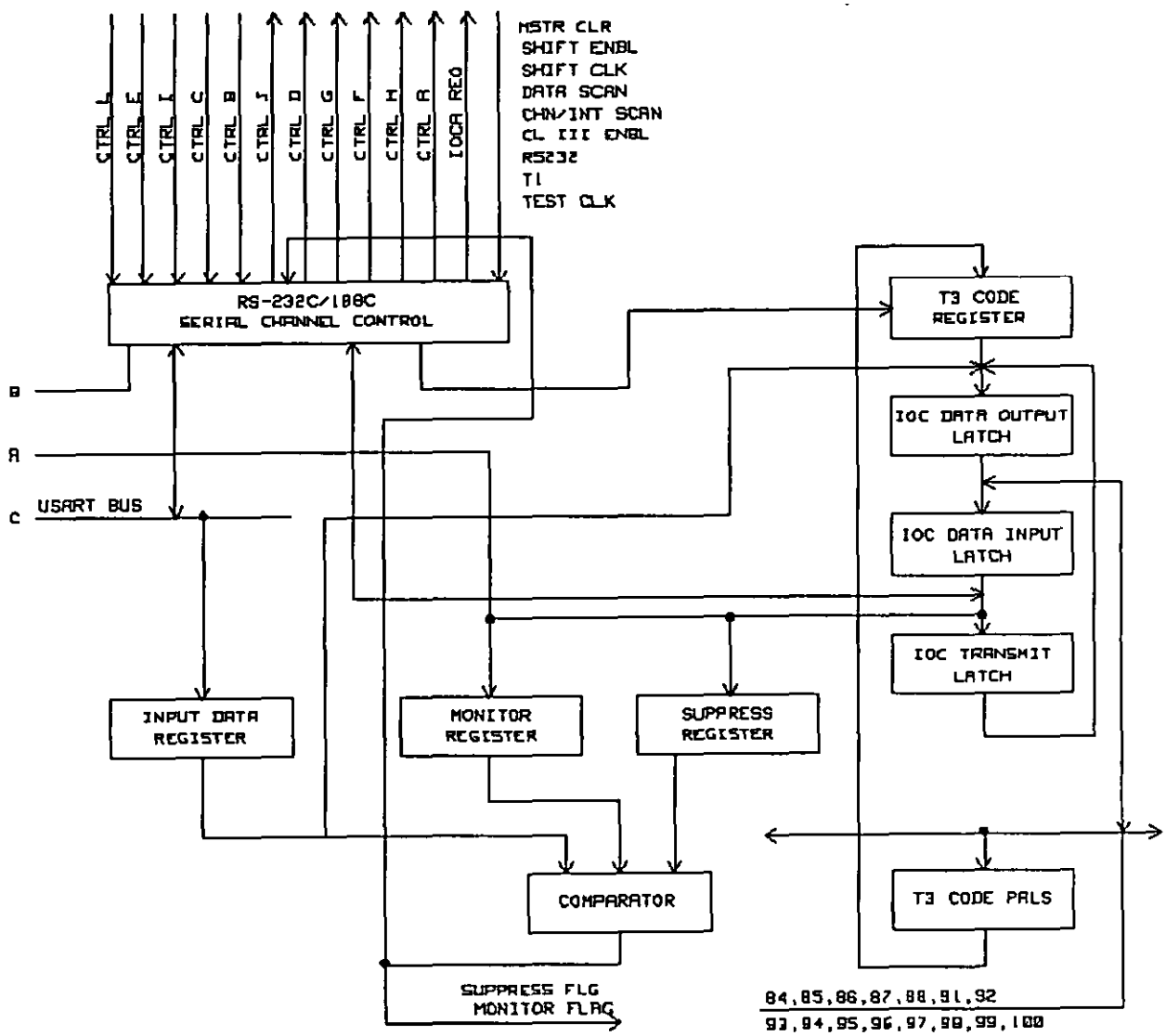


FIGURE 1. AEK functional block diagram - Continued.

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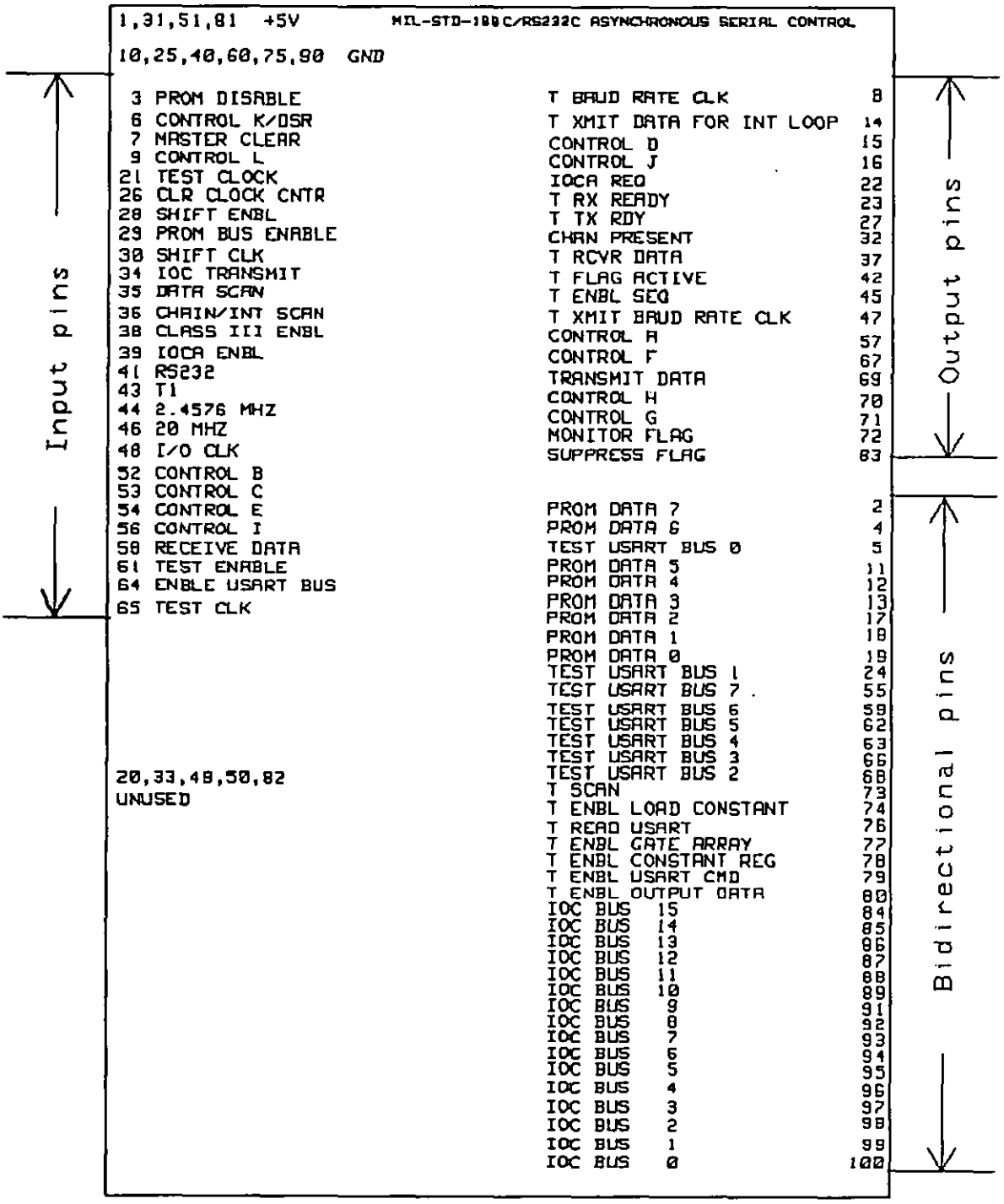


FIGURE 2. AEK pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

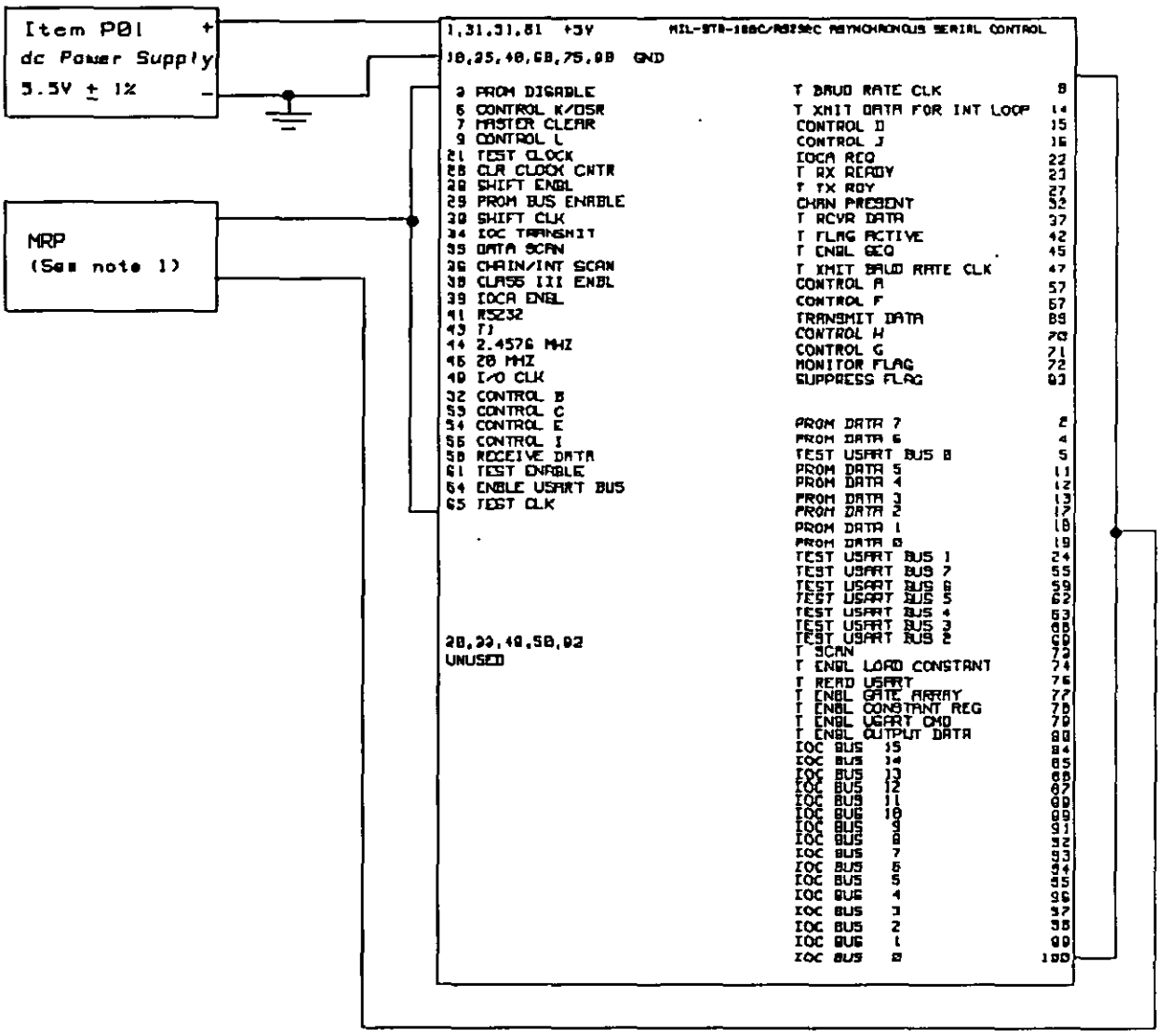
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AEK operating temperature and component temperature test circuit.

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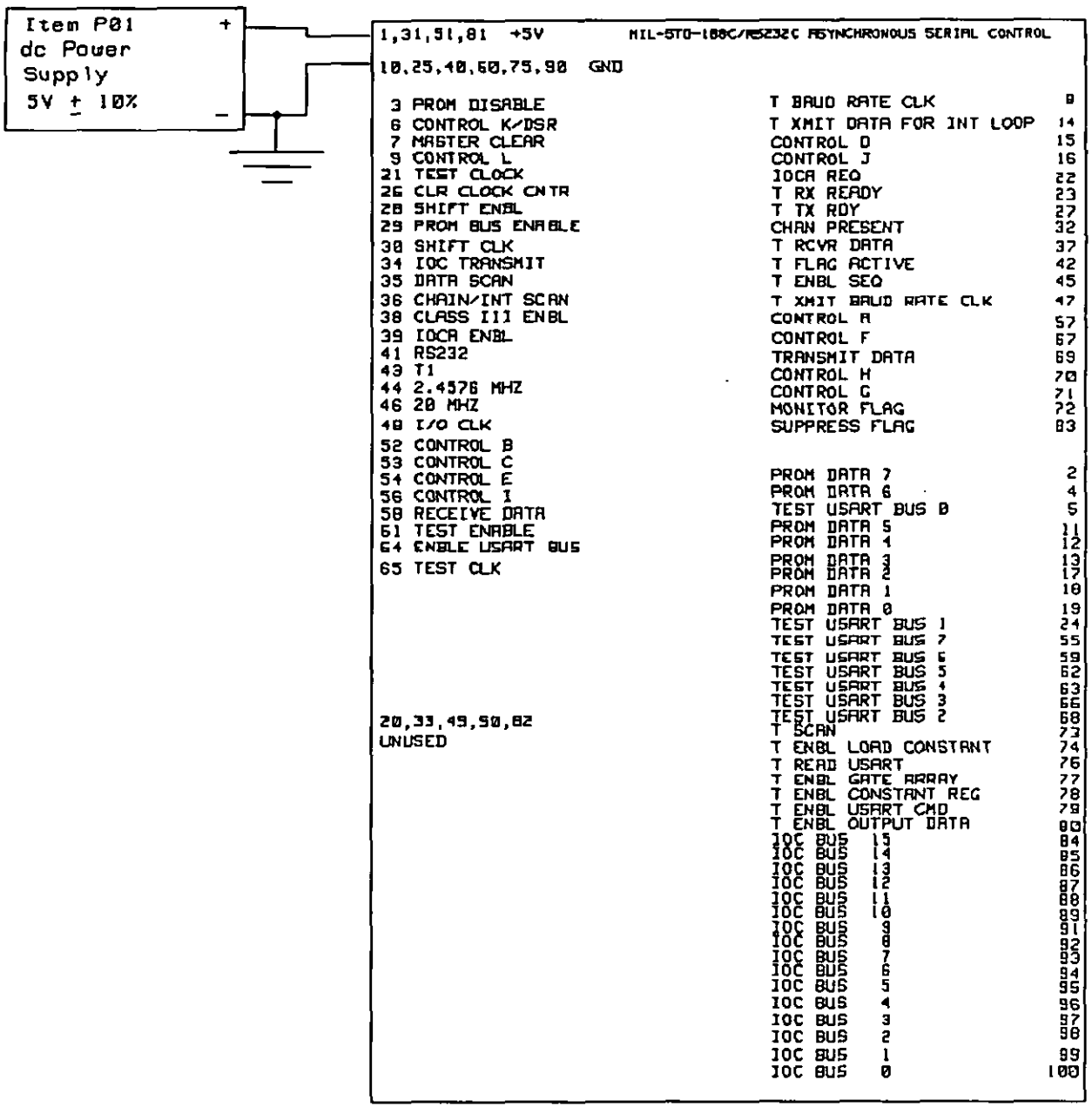


FIGURE 4. AEK vibration test circuit.

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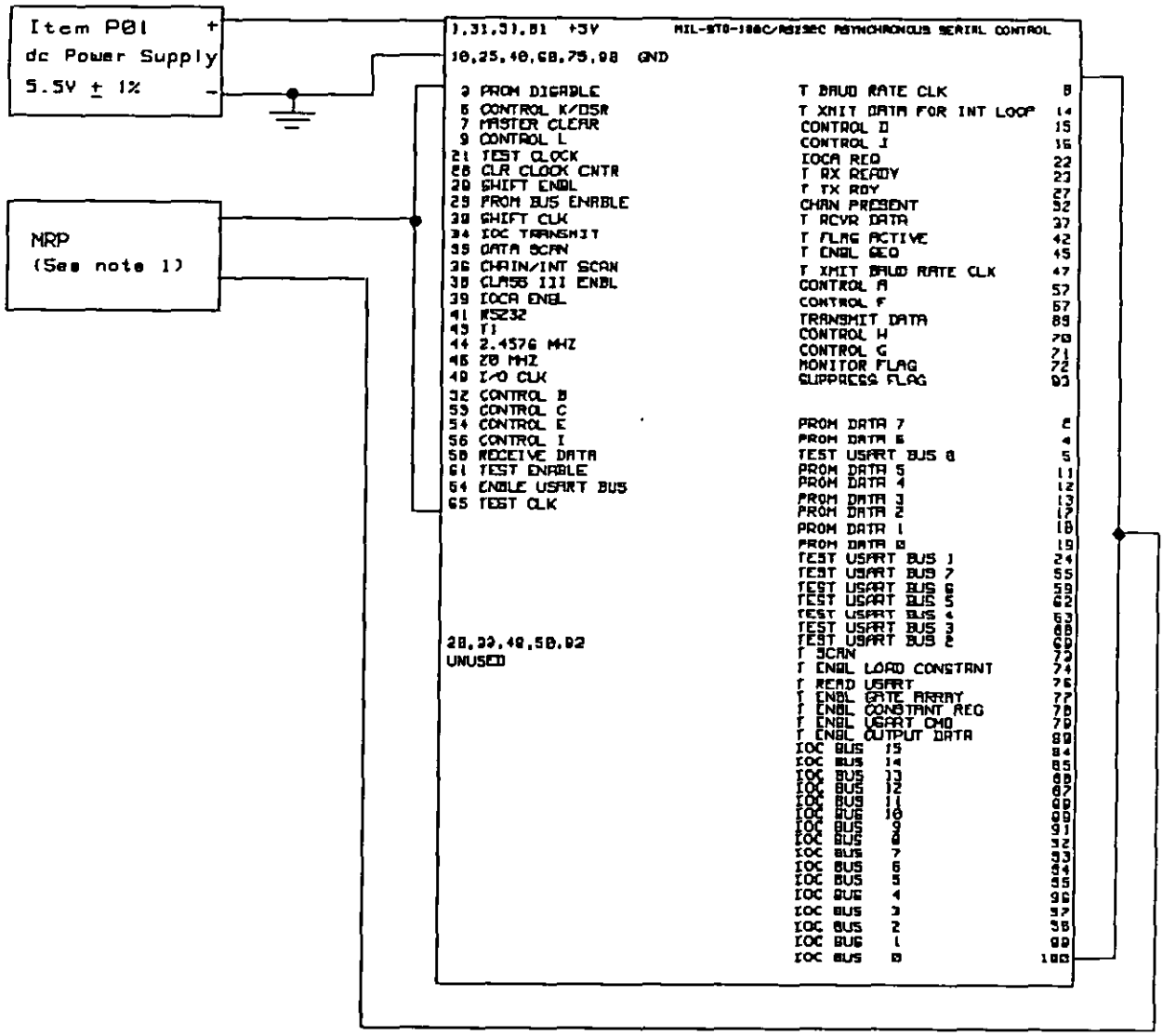
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AEK life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 8.25 W (max); 5.25 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PROM DISABLE	NA	0.4	2.4	-5.88	-2.84	1.21	175
CTRL K/DSR	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MSTR CLR	-1.5	0.4	2.4	-0.2	0.02	0.1	100
CTRL L	-1.5	0.4	2.4	-0.2	0.02	0.1	100
TEST CLK	-1.5	0.4	2.4	-0.4	0.02	0.1	75
CLR CLK CNTR	NA	0.4	2.4	-15.2	-4.11	1.27	100
SHIFT ENBL	NA	0.4	2.4	-5.63	-2.85	0.22	75
PROM ENBL	-1.5	0.4	2.4	-0.2	0.02	0.1	100
SHIFT CLK	NA	0.4	2.4	-7.43	-2.82	1.11	75
IOC XMIT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DATA SCAN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CHN/INT SCAN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CL III ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOCA ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RS232	-1.5	0.4	2.4	-0.2	0.02	0.1	125
T1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
2.4576	-1.2	0.4	2.4	-2.0	0.05	1.0	75
20 MHZ	-1.2	0.4	2.4	-2.0	0.05	1.0	75
I/O CLK	-1.2	0.4	2.4	-8.0	0.2	4.0	125
CTRL B	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CTRL C	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CTRL E	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CTRL I	-1.5	0.4	2.4	-0.2	0.02	0.1	75
RCV DATA	-1.5	0.4	2.4	-0.2	0.02	0.1	75
TEST ENBL	NA	0.4	2.4	-6.83	-2.75	0.72	200
ENBL USART	-1.5	0.4	2.4	-0.2	0.02	0.1	75
TEST CLK	-1.5	0.4	2.4	-0.25	0.03	1.0	75
PROM DATA 7	NA	NA	NA	NA	NA	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PROM DATA 6	NA	NA	NA	NA	NA	NA	75
TEST USART 0	NA	NA	NA	NA	NA	NA	100
PROM DATA 5	NA	NA	NA	NA	NA	NA	75
PROM DATA 4	NA	NA	NA	NA	NA	NA	75
PROM DATA 3	NA	NA	NA	NA	NA	NA	75
PROM DATA 2	NA	NA	NA	NA	NA	NA	75
PROM DATA 1	NA	NA	NA	NA	NA	NA	100
PROM DATA 0	NA	NA	NA	NA	NA	NA	100
TEST USART 1	NA	NA	NA	NA	NA	NA	100
TEST USART 7	NA	NA	NA	NA	NA	NA	75
TEST USART 6	NA	NA	NA	NA	NA	NA	75
TEST USART 5	NA	NA	NA	NA	NA	NA	75
TEST USART 4	NA	NA	NA	NA	NA	NA	75
TEST USART 3	NA	NA	NA	NA	NA	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
TEST USART 2	NA	NA	NA	NA	NA	NA	75
T SCAN	NA	NA	NA	NA	NA	NA	75
T ENBL CNST	NA	NA	NA	NA	NA	NA	75
T RD USART	NA	NA	NA	NA	NA	NA	75
T ENBL ARRAY	NA	NA	NA	NA	NA	NA	75
T ENBL REG	NA	NA	NA	NA	NA	NA	75
T ENBL USART	NA	NA	NA	NA	NA	NA	75
T ENBL DATA	NA	NA	NA	NA	NA	NA	75
IOC BUS 15	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 14	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 13	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 12	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 11	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 10	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC BUS 9	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 8	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 7	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 6	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 5	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 4	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 3	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 2	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 1	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 0	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
PROM DATA 7	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
PROM DATA 6	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TEST USART 0	2.4	0.5	-0.16	1.6	NA	NA	-40.0	100
BAUD RATE	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
PROM DATA 5	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
PROM DATA 4	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
PROM DATA 3	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T XMIT DATA	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
CTRL D	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CTRL J	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
PROM DATA 2	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
PROM DATA 1	2.4	0.5	-0.16	1.6	NA	NA	-40.0	100
PROM DATA 0	2.4	0.5	-0.16	1.6	NA	NA	-40.0	100
IOCA REQ	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
T RX RDY	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
TEST USART 1	2.4	0.5	-0.16	1.6	NA	NA	-40.0	100
T TX RDY	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
CHAN PRES	NA	0.5	NA	1.6	1.0	NA	NA	NA
T RCVR DATA	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
T FLAG ACT	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
T ENBL SEQ	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
T XMIT BAUD	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
TEST USART 7	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
CTRL A	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
TEST USART 6	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TEST USART 5	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TEST USART 4	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TEST USART 3	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CTRL F	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
TEST USART 2	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
XMIT DATA	NA	0.5	NA	1.6	NA	NA	-5.18	NA
CTRL H	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CTRL G	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MONITOR FLAG	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA
T SCAN	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T ENBL CNST	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T RD USART	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T ENBL ARRAY	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T ENBL REG	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T ENBL USART	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
T ENBL DATA	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
SUPPRESS FLG	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC BUS 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC BUS 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	th	I/O CLK to IOC BUS 15	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 14	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 13	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 12	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 11	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 10	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 9	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 8	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 7	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 6	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 5	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 4	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 3	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 2	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 1	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 0	47	47	47	-	-	-	nS	12
2	tsu	IOC BUS 7 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 6 to I/O CLK	24	24	24	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tsu	IOC BUS 5 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 4 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 3 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 2 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 1 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 0 to I/O CLK	24	24	24	-	-	-	nS	12
3	tsu	IOCA ENBL to I/O CLK	30	30	30	-	-	-	nS	12
3	tsu	IOC XMIT to I/O CLK	30	30	30	-	-	-	nS	12
4	tpd	20 MHZ to PROM DATA 7	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 6	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 5	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 4	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 3	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 2	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 1	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 0	-	-	-	250	250	250	nS	12
5	tpd	20 MHZ to T SCAN	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL CNST	-	-	-	128	128	128	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	20 MHZ to T RD USART	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL ARRAY	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL REG	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL USART	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL DATA	-	-	-	128	128	128	nS	12
6	tpd	20 MHZ to TEST USART 0	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 1	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 7	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 6	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 5	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 4	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 3	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 2	-	-	-	405	405	405	nS	12
7	tpd	DATA SCAN to IOCA REQ	-	-	-	96	96	96	nS	12
8	tpd	TEST ENBL to BAUD RATE	-	-	-	25	25	25	nS	12
8	tpd	TEST ENBL to T ENBL SEQ	-	-	-	25	25	25	nS	12
8	tpd	TEST ENBL to T XMIT BAUD	-	-	-	25	25	25	nS	12

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6.4 Module function. Both the MIL-STD-188C/RS232C/ Variable Character Length String (VACALES) synchronous serial control (AFK) and MIL-STD-188C/RS232C asynchronous serial control (AEK) modules direct serial Input/Output (I/O) operation for a single channel of a two-channel group. The following paragraphs describe both modules since their operation is similar. The synchronous module contains data latches, a Universal Synchronous/Asynchronous Receiver Transmitter (USART) and associated control Programmable Read-Only Memory (PROM), a control program sequencer, command register, source register, destination register and constant register, input and output data registers, a T3 code register and a serial channel control gate array. The asynchronous control module contains all of the logic listed for the synchronous module plus external suppress and monitor detection logic.

6.4.1 USART and control PROM. The USART for the asynchronous control module is a Programmable Communication Interface (PCI) integrated circuit that accepts parallel input data, formats and serializes the information based on its current operating mode and transmits the data in a serial bit stream. Simultaneously, serial data can be received, converted into parallel form, deformatted and sent to the Input/Output Controller (IOC) via the IOC bus. Operation of both USARTs is directed by a microprogram contained in a control PROM. The control PROM is a 2K (1024) x 8-bit memory with outputs sent to the source, destination and constant registers. The control program sequencer addresses the PROM.

6.4.2 Control program sequencer. The control program sequencer directs asynchronous I/O operations by addressing the USART control PROM for the routine required to handle the condition in effect. USART bus bits are applied to an encoder which directs the upper counter to the value 6 through 0 (excluding 4). The enable sequencer signal from the channel control gate array provides an up-count enable to the lower counter. The carryout function of the lower counter is applied to the counter for sequencer address bits 7-4 to allow sequential accessing through the microprogram. Address bits are applied to the control PROM.

6.4.3 Serial channel control gate array. The serial channel control gate array is an 84-pin leadless microcircuit designed to control serial I/O operations and interface the Input/Output Channel Adapter (IOCA) to the IOC.

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6.4.4 T3 code register. The T3 code register is an 8-bit register used to hold the T3 code from the serial channel control gate array. This code will be transmitted to the IOC at T3 time (via the IOC output data latch).

6.4.5 Data latches and transmit latch. The IOC output data latch, IOC input data latches and the IOC transmit latch are 16-bit, 16-bit and 8-bit latches, respectively. They are used to hold IOC input or output data. The IOC transmit latch receives data from the serial channel control gate array for transmission to the IOC.

6.4.6 Input and output data registers. The input data registers and output data register transfer data to and from the USART bus. The output data register is enabled by the source register bit 0 and clocked by the serial channel control gate array. The input data register is clocked by destination register bit 7 and enabled by the serial channel control gate array.

6.4.7 Command register. The USART command register is loaded by destination register bit 6 and its output enabled by source register bit 1. The command register is loaded from the serial channel control gate array or constant register via the USART bus, with outputs going to the control sequencer and being returned to the USART bus.

6.4.8 Baud rate generator (asynchronous control module only). The baud rate generator is used to provide the clock input to the USART. Baud rate is program selectable via the serial mode word stored in I/O control memory location A and the serial channel control gate array.

6.4.9 Source, destination and constant registers. The source, destination and constant registers are 8-bit registers used to hold control PROM outputs. The contents of odd control PROM addresses are sent to the source register. The contents of even control PROM addresses are sent to the destination register or to the constant register depending upon the condition of source register bit 6. The constant register holds an 8-bit constant that is applied to the USART bus.

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6.4.10 Monitor register. The monitor register is an 8-bit register that holds a character to be compared against an incoming bit stream. When the received character matches the one in the monitor register, the monitor flag to the serial control gate array is set. The monitor register is loaded from the IOCA bus by the serial channel control gate array as a result of an IOC execution of a load control memory macroinstruction (I/O control memory location 9).

6.4.11 Suppress register. The suppress register is an 8-bit register that holds a character which is to be ignored by the IOC. When a character in an incoming bit stream matches the contents of the suppress register, the suppress flag is generated to the serial channel control gate array and that character is ignored. The suppress register is loaded from the IOCA bus by the serial channel control gate array as the result of an IOC execution of the load control memory macroinstruction (I/O control memory location 8).

6.4.12 Monitor and suppress comparators. The monitor and suppress comparators are 8-bit comparators that are enabled by the serial channel control gate array as the result of an IOC execution of a set monitor or set suppress (chaining) macroinstruction. The outputs from the comparators are the monitor and suppress flags.

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6.4.13 Description of I/O pins.

Signal name	Pin number	I/O	Description
PROM DISABLE	3	I	PROM DISABLE
CTRL K/DSR	6	I	CONTROL K/DSR
MSTR CLR	7	I	MASTER CLEAR
CTRL L	9	I	CONTROL L
TEST CLK	21	I	TEST CLOCK
CLR CLK CNTR	26	I	CLR CLOCK CTR
SHIFT ENBL	28	I	SHIFT ENBL
PROM ENBL	29	I	PROM BUS ENABLE
SHIFT CLK	30	I	SHIFT CLK
IOC XMIT	34	I	IOC TRANSMIT
DATA SCAN	35	I	DATA SCAN
CHN/INT SCAN	36	I	CHAIN/INT SCAN
CL III ENBL	38	I	CLASS III ENBL
IOCA ENBL	39	I	IOCA ENBL
RS232	41	I	RS232
T1	43	I	T1
2.4576 MHZ	44	I	2.4576 MHZ
20 MHZ	46	I	20 MHZ
I/O CLK	48	I	I/O CLK
CTRL B	52	I	CONTROL B

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6.4.13 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CTRL C	53	I	CONTROL C
CTRL E	54	I	CONTROL E
CTRL I	56	I	CONTROL I
RCV DATA	58	I	RECEIVE DATA
TEST ENBL	61	I	TEST ENABLE
ENBL USART	64	I	ENABLE USART BUS
TEST CLK	65	I	TEST CLK
PROM DATA 7	2	I/O	PROM DATA 7
PROM DATA 6	4	I/O	PROM DATA 6
TEST USART 0	5	I/O	TEST USART BUS 0
BAUD RATE	8	O	T BAUD RATE CLK
PROM DATA 5	11	I/O	PROM DATA 5
PROM DATA 4	12	I/O	PROM DATA 4
PROM DATA 3	13	I/O	PROM DATA 3
T XMIT DATA	14	O	T XMIT DATA FOR INT LOOP
CTRL D	15	O	CONTROL D
CTRL J	16	O	CONTROL J
PROM DATA 2	17	I/O	PROM DATA 2
PROM DATA 1	18	I/O	PROM DATA 1
PROM DATA 0	19	I/O	PROM DATA 0

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6.4.13 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOCA REQ	22	O	IOCA REQ
T RX RDY	23	O	T RX RDY
TEST USART 1	24	I/O	TEST USART BUS 1
T TX RDY	27	O	T TX RDY
CHAN PRES	32	O	CHAN PRESENT
T RCVR DATA	37	O	T RCVR DATA
T FLAG ACT	42	O	T FLAG ACTIVE
T ENBL SEQ	45	O	T ENBL SEQ
T XMIT BAUD	47	O	T XMIT BAUD RATE CLK
TEST USART 7	55	I/O	TEST USART BUS 7
CTRL A	57	O	CONTROL A
TEST USART 6	59	I/O	TEST USART BUS 6
TEST USART 5	62	I/O	TEST USART BUS 5
TEST USART 4	63	I/O	TEST USART BUS 4
TEST USART 3	66	I/O	TEST USART BUS 3
CTRL F	67	O	CONTROL F
TEST USART 2	68	I/O	TEST USART BUS 2
XMIT DATA	69	O	TRANSMIT DATA
CTRL H	70	O	CONTROL H
CTRL G	71	O	CONTROL G

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6.4.13 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MONITOR FLAG	72	I/O	MONITOR FLAG
T SCAN	73	I/O	T SCAN
T ENBL CNST	74	I/O	T ENBL LOAD CONSTANT
T RD USART	76	I/O	T READ USART
T ENBL ARRAY	77	I/O	T ENBL GATE ARRAY
T ENBL REG	78	I/O	T ENBL CONSTANT REG
T ENBL USART	79	I/O	T ENBL USART CMD
T ENBL DATA	80	I/O	T ENBL OUTPUT DATA
SUPPRESS FLG	83	O	SUPPRESS FLAG
IOC BUS 15	84	I/O	IOC BUS 15
IOC BUS 14	85	I/O	IOC BUS 14
IOC BUS 13	86	I/O	IOC BUS 13
IOC BUS 12	87	I/O	IOC BUS 12
IOC BUS 11	88	I/O	IOC BUS 11
IOC BUS 10	89	I/O	IOC BUS 10
IOC BUS 9	91	I/O	IOC BUS 9
IOC BUS 8	92	I/O	IOC BUS 8
IOC BUS 7	93	I/O	IOC BUS 7
IOC BUS 6	94	I/O	IOC BUS 6
IOC BUS 5	95	I/O	IOC BUS 5

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6.4.13 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC BUS 4	96	I/O	IOC BUS 4
IOC BUS 3	97	I/O	IOC BUS 3
IOC BUS 2	98	I/O	IOC BUS 2
IOC BUS 1	99	I/O	IOC BUS 1
IOC BUS 0	100	I/O	IOC BUS 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N737)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,20, 33,49, 50,82	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	20,33,49, 50,82				
A3	Isolation, unused to unused pins	20	33,49,50, 82				
A4	Isolation, unused to unused pins	33	49,50,82				
A5	Isolation, unused to unused pins	49	50,82				
A6	Isolation, unused to unused pins	50	82	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A6
Isolation measurement	NA	± 500 KOhm	A1-A6

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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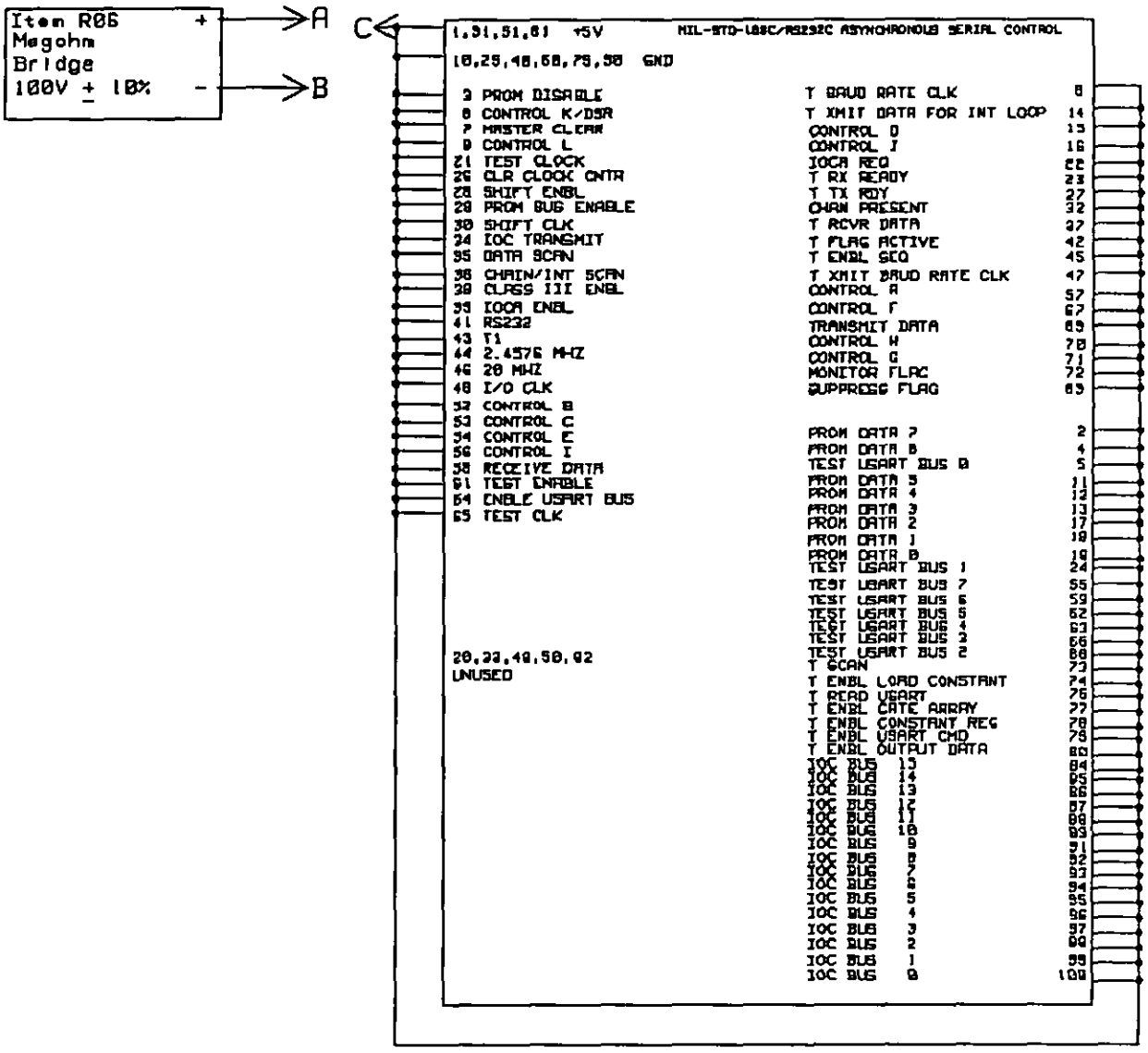


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, GND	10	60				
B6	Continuity, Vcc	1	31				
B7	Continuity, Vcc	51	81				
B8	Continuity, Vcc	1	51	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B8

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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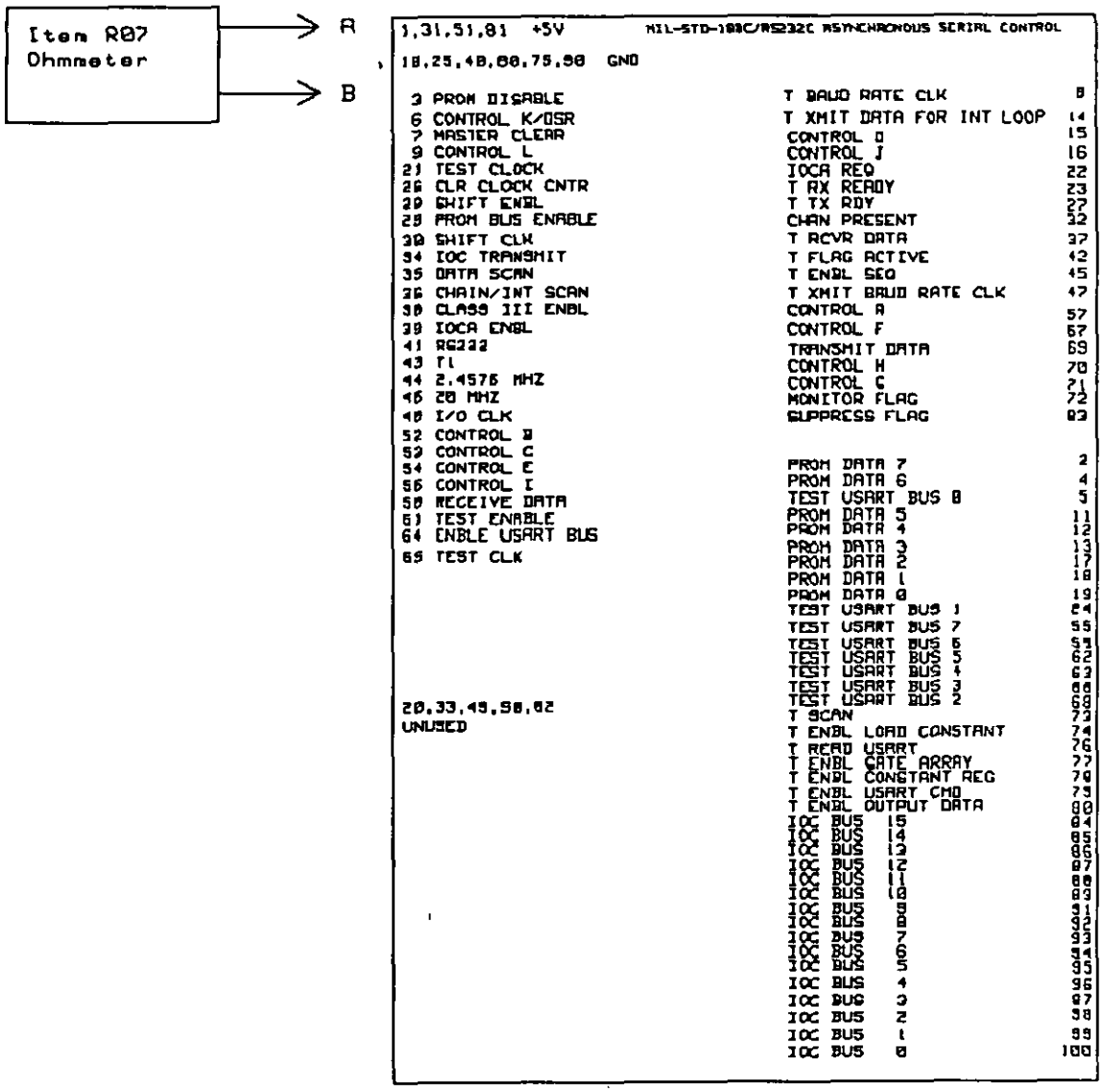


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all re-maining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.7 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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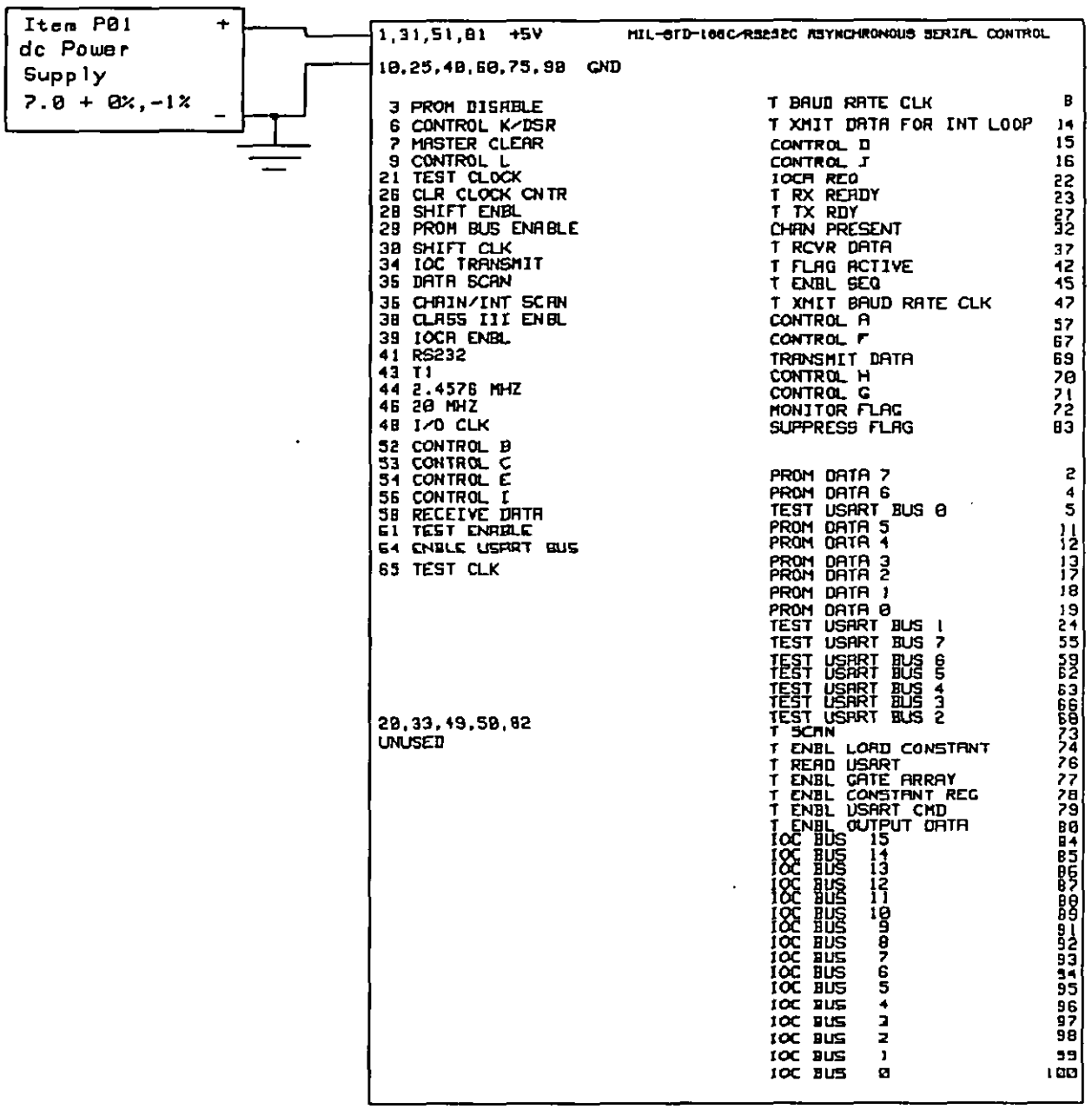


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	PROM DISABLE	3	5.5	-0.12	1.21	mA	1-3
D2	CTRL K/DSR	6	5.5	0.0	0.1	mA	1-3
D3	MSTR CLR	7					
D4	CTRL L	9	5.5	0.0	0.1	mA	1-3
D5	CLR CLK CNTR	26	5.5	-0.17	1.27	mA	1-3
D6	SHIFT ENBL	28	5.5	-0.12	0.22	mA	1-3
D7	PROM ENBL	29	5.5	0.0	0.1	mA	1-3
D8	SHIFT CLK	30	5.5	-0.12	1.11	mA	1-3
D9	IOC XMIT	34	5.5	0.0	0.1	mA	1-3
D10	DATA SCAN	35					
D11	CHN/INT SCAN	36					
D12	CL III ENBL	38					
D13	IOCA ENBL	39					
D14	RS232	41					
D15	T1	43	5.5	0.0	0.1	mA	1-3
D16	2.4576 MHZ	44	5.5	0.0	1.0	mA	1-3
D17	20 MHZ	46	5.5	0.0	1.0	mA	1-3
D18	I/O CLK	48	5.5	0.0	4.0	mA	1-3
D19	CTRL B	52	5.5	0.0	0.1	mA	1-3
D20	CTRL C	53	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	CTRL E	54	5.5	0.0	0.1	mA	1-3
D22	CTRL I	56					
D23	RCV DATA	58	5.5	0.0	0.1	mA	1-3
D24	TEST ENBL	61	5.5	-0.12	0.72	mA	1-3
D25	ENBL USART	64	5.5	0.0	0.1	mA	1-3
D26	TEST CLK	65	5.5	0.0	1.0	mA	1-3
D27	TEST CLK	21	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D27
Power supply current limit	2.25 A	NA	D1-D27
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D27
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D27 D1-D27
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 66.5 uA ± 5.0 uA ± 72.0 uA ± 17.0 uA ± 5.0 uA ± 61.5 uA ± 5.0 uA ± 50.0 uA ± 200.0 uA ± 5.0 uA ± 42.0 uA ± 5.0 uA ± 50.0 uA ± 5.0 uA	D1 D2-D4 D5 D6 D7 D8 D9-D15 D16,D17 D18 D19-D23 D24 D25 D26 D27

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E1	PROM DATA 7	2	0.0	-225.0	-40.0	mA	1-4
E2	PROM DATA 6	4					
E3	TEST USART 0	5	0.0	-225.0	-40.0	mA	1-4
E4	BAUD RATE	8	0.0	-260.0	-30.0	mA	1-4
E5	PROM DATA 5	11	0.0	-225.0	-40.0	mA	1-4
E6	PROM DATA 4	12					
E7	PROM DATA 3	13	0.0	-225.0	-40.0	mA	1-4
E8	T XMIT DATA	14	0.0	-260.0	-30.0	mA	1-4
E9	CTRL D	15					
E10	CTRL J	16	0.0	-260.0	-30.0	mA	1-4
E11	PROM DATA 2	17	0.0	-225.0	-40.0	mA	1-4
E12	PROM DATA 1	18					
E13	PROM DATA 0	19	0.0	-225.0	-40.0	mA	1-4
E14	IOCA REQ	22	0.0	-260.0	-15.0	mA	1-4
E15	T RX RDY	23	0.0	-260.0	-30.0	mA	1-4
E16	TEST USART 1	24	0.0	-225.0	-40.0	mA	1-4
E17	T TX RDY	27	0.0	-260.0	-30.0	mA	1-4
E18	T RCVR DATA	37					
E19	T FLAG ACT	42					
E20	T ENBL SEQ	45	0.0	-260.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	T XMIT BAUD	47	0.0	-260.0	-30.0	mA	1-4
E22	TEST USART 7	55	0.0	-225.0	-40.0	mA	1-4
E23	CTRL A	57	0.0	-260.0	-30.0	mA	1-4
E24	TEST USART 6	59	0.0	-225.0	-40.0	mA	1-4
E25	TEST USART 5	62					
E26	TEST USART 4	63					
E27	TEST USART 3	66	0.0	-225.0	-40.0	mA	1-4
E28	CTRL F	67	0.0	-260.0	-30.0	mA	1-4
E29	TEST USART 2	68	0.0	-225.0	-40.0	mA	1-4
E30	XMIT DATA	69	0.0	-7.84	-5.18	mA	1-4
E31	CTRL H	70	0.0	-260.0	-30.0	mA	1-4
E32	CTRL G	71	0.0	-260.0	-30.0	mA	1-4
E33	MONITOR FLAG	72	0.0	-85.0	-15.0	mA	1-4
E34	T SCAN	73	0.0	-225.0	-40.0	mA	1-4
E35	T ENBL CNST	74					
E36	T RD USART	76					
E37	T ENBL ARRAY	77					
E38	T ENBL REG	78					
E39	T ENBL USART	79					
E40	T ENBL DATA	80	0.0	-225.0	-40.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E41	SUPPRESS FLG	83	0.0	-85.0	-15.0	mA	1-4
E42	IOC BUS 15	84	0.0	-150.0	-60.0	mA	1-4
E43	IOC BUS 14	85					
E44	IOC BUS 13	86					
E45	IOC BUS 12	87					
E46	IOC BUS 11	88					
E47	IOC BUS 10	89					
E48	IOC BUS 9	91					
E49	IOC BUS 8	92					
E50	IOC BUS 7	93					
E51	IOC BUS 6	94					
E52	IOC BUS 5	95					
E53	IOC BUS 4	96					
E54	IOC BUS 3	97					
E55	IOC BUS 2	98					
E56	IOC BUS 1	99					
E57	IOC BUS 0	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E57
Power supply current limit	2.25 A	NA	E1-E57
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E57
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E57 E1-E57
Output current measurement	NA	± 9.25 mA	E1-E3
	NA	± 11.5 mA	E4
	NA	± 9.25 mA	E5-E7
	NA	± 11.5 mA	E8-E10
	NA	± 9.25 mA	E11-E13
	NA	± 12.2 mA	E14
	NA	± 11.5 mA	E15
	NA	± 9.25 mA	E16
	NA	± 11.5 mA	E17-E21
	NA	± 9.25 mA	E22
	NA	± 11.5 mA	E23
	NA	± 9.25 mA	E24-E27
	NA	± 11.5 mA	E28
	NA	± 9.25 mA	E29
	NA	± 0.13 mA	E30
	NA	± 11.5 mA	E31,E32
	NA	± 3.5 mA	E33
	NA	± 9.25 mA	E34-E40
	NA	± 3.5 mA	E41
	NA	± 4.5 mA	E42-E57

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	PROM DISABLE	3	2.4	-5.88	-2.84	mA	1-3
F2	CTRL K/DSR	6	2.4	-0.2	0.02	mA	1-3
F3	MSTR CLR	7					
F4	CTRL L	9	2.4	-0.2	0.02	mA	1-3
F5	CLR CLK CNTR	26	2.4	-15.2	-4.11	mA	1-3
F6	SHIFT ENBL	28	2.4	-5.63	-2.85	mA	1-3
F7	PROM ENBL	29	2.4	-0.2	0.02	mA	1-3
F8	SHIFT CLK	30	2.4	-7.43	-2.82	mA	1-3
F9	IOC XMIT	34	2.4	-0.6	0.02	mA	1-3
F10	DATA SCAN	35					
F11	CHN/INT SCAN	36					
F12	CL III ENBL	38					
F13	IOCA ENBL	39	2.4	-0.6	0.02	mA	1-3
F14	RS232	41	2.4	-0.2	0.02	mA	1-3
F15	T1	43	2.4	-0.6	0.02	mA	1-3
F16	2.4576 MHZ	44	2.4	-2.0	0.05	mA	1-3
F17	20 MHZ	46	2.4	-2.0	0.05	mA	1-3
F18	I/O CLK	48	2.4	-8.0	0.2	mA	1-3
F19	CTRL B	52	2.4	-0.2	0.02	mA	1-3
F20	CTRL C	53	2.4	-0.2	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F21	CTRL E	54	2.4	-0.2	0.02	mA	1-3
F22	CTRL I	56					
F23	RCV DATA	58	2.4	-0.2	0.02	mA	1-3
F24	TEST ENBL	61	2.4	-6.83	-2.75	mA	1-3
F25	ENBL USART	64	2.4	-0.2	0.02	mA	1-3
F26	TEST CLK	65	2.4	-0.25	0.03	mA	1-3
F27	IOC BUS 15	84	2.4	-0.25	0.07	mA	1-3
F28	IOC BUS 14	85					
F29	IOC BUS 13	86					
F30	IOC BUS 12	87					
F31	IOC BUS 11	88					
F32	IOC BUS 10	89					
F33	IOC BUS 9	91					
F34	IOC BUS 8	92					
F35	IOC BUS 7	93					
F36	IOC BUS 6	94					
F37	IOC BUS 5	95					
F38	IOC BUS 4	96					
F39	IOC BUS 3	97					
F40	IOC BUS 2	98	2.4	-0.25	0.07	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F41	IOC BUS 1	.99	2.4	-0.25	0.07	mA	1-3
F42	IOC BUS 0	100	2.4	-0.25	0.07	mA	1-3
F43	TEST CLK	21	2.4	-0.4	0.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:30Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F43
Power supply current limit	2.25 A	NA	F1-F43
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F43
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F43
low, (VL)	0.0 V	± 100 mV	F1-F43
Input current measurement	NA	± 152 uA	F1
	NA	± 11 uA	F2-F4
	NA	± 550 uA	F5
	NA	± 139 uA	F6
	NA	± 11 uA	F7
	NA	± 230 uA	F8
	NA	± 31 uA	F9-F13
	NA	± 11 uA	F14
	NA	± 31 uA	F15
	NA	± 102 uA	F16,F17
	NA	± 410 uA	F18
	NA	± 11 uA	F19-F23
	NA	± 204 uA	F24
	NA	± 11 uA	F25
	NA	± 14 uA	F26
	NA	± 16 uA	F27-F42
	NA	± 21 uA	F43

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	PROM DISABLE	3	0.4	-5.88	-2.84	mA	1-3
G2	CTRL K/DSR	6	0.4	-0.2	0.02	mA	1-3
G3	MSTR CLR	7					
G4	CTRL L	9	0.4	-0.2	0.02	mA	1-3
G5	CLR CLK CNTR	26	0.4	-15.2	-4.11	mA	1-3
G6	SHIFT ENBL	28	0.4	-5.63	-2.85	mA	1-3
G7	PROM ENBL	29	0.4	-0.2	0.02	mA	1-3
G8	SHIFT CLK	30	0.4	-7.43	-2.82	mA	1-3
G9	IOC XMIT	34	0.4	-0.6	0.02	mA	1-3
G10	DATA SCAN	35					
G11	CHN/INT SCAN	36					
G12	CL III ENBL	38					
G13	IOCA ENBL	39	0.4	-0.6	0.02	mA	1-3
G14	RS232	41	0.4	-0.2	0.02	mA	1-3
G15	T1	43	0.4	-0.6	0.02	mA	1-3
G16	2.4576 MHZ	44	0.4	-2.0	0.05	mA	1-3
G17	20 MHZ	46	0.4	-2.0	0.05	mA	1-3
G18	I/O CLK	48	0.4	-8.0	0.2	mA	1-3
G19	CTRL B	52	0.4	-0.2	0.02	mA	1-3
G20	CTRL C	53	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	CTRL E	54	0.4	-0.2	0.02	mA	1-3
G22	CTRL I	56					
G23	RCV DATA	58	0.4	-0.2	0.02	mA	1-3
G24	TEST ENBL	61	0.4	-6.83	-2.75	mA	1-3
G25	ENBL USART	64	0.4	-0.2	0.02	mA	1-3
G26	TEST CLK	65	0.4	-0.25	0.03	mA	1-3
G27	IOC BUS 15	84	0.4	-0.25	0.07	mA	1-3
G28	IOC BUS 14	85					
G29	IOC BUS 13	86					
G30	IOC BUS 12	87					
G31	IOC BUS 11	88					
G32	IOC BUS 10	89					
G33	IOC BUS 9	91					
G34	IOC BUS 8	92					
G35	IOC BUS 7	93					
G36	IOC BUS 6	94					
G37	IOC BUS 5	95					
G38	IOC BUS 4	96					
G39	IOC BUS 3	97					
G40	IOC BUS 2	98	0.4	-0.25	0.07	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	IOC BUS 1	99	0.4	-0.25	0.07	mA	1-3
G42	IOC BUS 0	100	0.4	-0.25	0.07	mA	1-3
G43	TEST CLK	21	0.4	-0.4	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G43
Power supply current limit	2.25 A	NA	G1-G43
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G43
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G43
low, (VL)	0.0 V	± 100 mV	G1-G43
Input current measurement	NA	± 152 uA	G1
	NA	± 11 uA	G2-G4
	NA	± 550 uA	G5
	NA	± 139 uA	G6
	NA	± 11 uA	G7
	NA	± 230 uA	G8
	NA	± 31 uA	G9-G13
	NA	± 11 uA	G14
	NA	± 31 uA	G15
	NA	± 102 uA	G16,G17
	NA	± 410 uA	G18
	NA	± 11 uA	G19-G23
	NA	± 204 uA	G24
	NA	± 11 uA	G25
	NA	± 14 uA	G26
	NA	± 16 uA	G27-G42
	NA	± 21 uA	G43

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	PROM DATA 7	2	-0.16	2.4	4.5	V	1-4
H2	PROM DATA 6	4					
H3	TEST USART 0	5					
H4	BAUD RATE	8					
H5	PROM DATA 5	11					
H6	PROM DATA 4	12					
H7	PROM DATA 3	13					
H8	T XMIT DATA	14					
H9	CTRL D	15					
H10	CTRL J	16					
H11	PROM DATA 2	17					
H12	PROM DATA 1	18					
H13	PROM DATA 0	19					
H14	IOCA REQ	22					
H15	T RX RDY	23					
H16	TEST USART 1	24					
H17	T TX RDY	27					
H18	T RCVR DATA	37					
H19	T FLAG ACT	42					
H20	T ENBL SEQ	45	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	T XMIT BAUD	47	-0.16	2.4	4.5	V	1-4
H22	TEST USART 7	55					
H23	CTRL A	57					
H24	TEST USART 6	59					
H25	TEST USART 5	62					
H26	TEST USART 4	63					
H27	TEST USART 3	66					
H28	CTRL F	67					
H29	TEST USART 2	68					
H30	CTRL H	70					
H31	CTRL G	71					
H32	MONITOR FLAG	72					
H33	T SCAN	73					
H34	T ENBL CNST	74					
H35	T RD USART	76					
H36	T ENBL ARRAY	77					
H37	T ENBL REG	78					
H38	T ENBL USART	79					
H39	T ENBL DATA	80					
H40	SUPPRESS FLG	83	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
H41	IOC BUS 15	84	-0.4	2.4	4.5	V	1-4
H42	IOC BUS 14	85					
H43	IOC BUS 13	86					
H44	IOC BUS 12	87					
H45	IOC BUS 11	88					
H46	IOC BUS 10	89					
H47	IOC BUS 9	91					
H48	IOC BUS 8	92					
H49	IOC BUS 7	93					
H50	IOC BUS 6	94					
H51	IOC BUS 5	95					
H52	IOC BUS 4	96					
H53	IOC BUS 3	97					
H54	IOC BUS 2	98					
H55	IOC BUS 1	99					
H56	IOC BUS 0	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H56
Power supply current limit	2.25 A	NA	H1-H56
Forcing current, (Ioh)	-0.16 mA -0.4 mA	± 16 uA ± 40 uA	H1-H40 H41-H56
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H56 H1-H56
Output voltage measurement	NA	± 105 mV	H1-H56

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

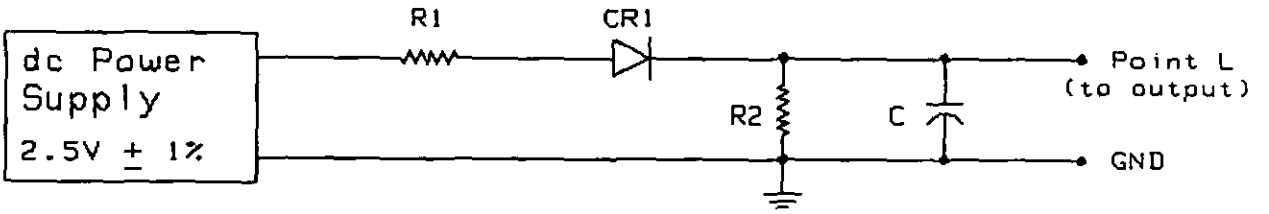
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,4,5,8,11,12,13, 14,15,16,17,18, 19,22,23,24,27, 37,42,45,47,55, 57,59,62,63,66, 67,68,70,71,72, 73,74,76,77,78, 79,80,83	16000	1.6	0.5
32,69	NA	1.6	0.5
84,85,86,87,88, 89,91,92,93,94, 95,96,97,98,99, 100	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	PROM DATA 7	2	1.6	0.0	0.5	V	1-4
J2	PROM DATA 6	4					
J3	TEST USART 0	5					
J4	BAUD RATE	8					
J5	PROM DATA 5	11					
J6	PROM DATA 4	12					
J7	PROM DATA 3	13					
J8	T XMIT DATA	14					
J9	CTRL D	15					
J10	CTRL J	16					
J11	PROM DATA 2	17					
J12	PROM DATA 1	18					
J13	PROM DATA 0	19					
J14	IOCA REQ	22					
J15	T RX RDY	23					
J16	TEST USART 1	24					
J17	T TX RDY	27					
J18	CHAN PRES	32					
J19	T RCVR DATA	37					
J20	T FLAG ACT	42	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	T ENBL SEQ	45	1.6	0.0	0.5	V	1-4
J22	T XMIT BAUD	47					
J23	TEST USART 7	55					
J24	CTRL A	57					
J25	TEST USART 6	59					
J26	TEST USART 5	62					
J27	TEST USART 4	63					
J28	TEST USART 3	66					
J29	CTRL F	67					
J30	TEST USART 2	68					
J31	XMIT DATA	69					
J32	CTRL H	70					
J33	CTRL G	71					
J34	MONITOR FLAG	72					
J35	T SCAN	73					
J36	T ENBL CNST	74					
J37	T RD USART	76					
J38	T ENBL ARRAY	77					
J39	T ENBL REG	78					
J40	T ENBL USART	79	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J41	T ENBL DATA	80	1.6	0.0	0.5	V	1-4
J42	SUPPRESS FLG	83	1.6	0.0	0.5	V	1-4
J43	IOC BUS 15	84	6.0	0.0	0.5	V	1-4
J44	IOC BUS 14	85					
J45	IOC BUS 13	86					
J46	IOC BUS 12	87					
J47	IOC BUS 11	88					
J48	IOC BUS 10	89					
J49	IOC BUS 9	91					
J50	IOC BUS 8	92					
J51	IOC BUS 7	93					
J52	IOC BUS 6	94					
J53	IOC BUS 5	95					
J54	IOC BUS 4	96					
J55	IOC BUS 3	97					
J56	IOC BUS 2	98					
J57	IOC BUS 1	99					
J58	IOC BUS 0	100	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:30Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J58
Power supply current limit	2.25 A	NA	J1-J58
Forcing current, (Iol)	1.6 mA 6.0 mA	± 160 uA ± 600 uA	J1-J42 J43-J58
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J58 J1-J58
Output voltage measurement	NA	± 25 mV	J1-J58

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:30Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	BAUD RATE	8	2.4	-0.02	0.02	mA	1-3
K2	T XMIT DATA	14					
K3	T RX RDY	23					
K4	T TX RDY	27	2.4	-0.02	0.02	mA	1-3
K5	CHAN PRES	32	5.5	-1.0	1.0	mA	1-3
K6	T RCVR DATA	37	2.4	-0.02	0.02	mA	1-3
K7	T FLAG ACT	42					
K8	T ENBL SEQ	45					
K9	T XMIT BAUD	47	2.4	-0.02	0.02	mA	1-3
Low level							
K10	BAUD RATE	8	0.4	-0.02	0.02	mA	1-3
K11	T XMIT DATA	14					
K12	T RX RDY	23					
K13	T TX RDY	27					
K14	T RCVR DATA	37					
K15	T FLAG ACT	42					
K16	T ENBL SEQ	45					
K17	T XMIT BAUD	47	0.4	-0.02	0.02	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K17
Power supply current limit	2.25 A	NA	K1-K17
Forcing voltage,			
(Voh)	2.4 V	± 50 mV	K1-K4
(Voh)	5.5 V	± 50 mV	K5
(Voh)	2.4 V	± 50 mV	K6-K9
(Vol)	0.4 V	± 50 mV	K10-K17
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	K1-K17
low, (VL)	0.0 V	± 100 mV	K1-K17
Output current measurement	NA	± 2 uA	K1-K4
	NA	± 100 uA	K5
	NA	± 2 uA	K6-K9
	NA	± 2 uA	K10-K17

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CTRL K/DSR	6	-18.0	-1.5	0.0	V	1-3
L2	MSTR CLR	7					
L3	CTRL L	9					
L4	PROM ENBL	29	-18.0	-1.5	0.0	V	1-3
L5	IOC XMIT	34	-18.0	-1.2	0.0	V	1-3
L6	DATA SCAN	35					
L7	CHN/INT SCAN	36					
L8	CL III ENBL	38					
L9	IOCA ENBL	39	-18.0	-1.2	0.0	V	1-3
L10	RS232	41	-18.0	-1.5	0.0	V	1-3
L11	T1	43	-18.0	-1.2	0.0	V	1-3
L12	2.4576 MHZ	44					
L13	20 MHZ	46					
L14	I/O CLK	48	-18.0	-1.2	0.0	V	1-3
L15	CTRL B	52	-18.0	-1.5	0.0	V	1-3
L16	CTRL C	53					
L17	CTRL E	54					
L18	CTRL I	56					
L19	RCV DATA	58					
L20	ENBL USART	64	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	TEST CLK	65	-18.0	-1.5	0.0	V	1-3
L22	TEST CLK	21	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L22
Power supply current limit	NA	NA	L1-L22
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L22
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L22 L1-L22
Input voltage measurement	NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1-L4 L5-L9 L10 L11-L14 L15-L22

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1	5.4	10.0	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 230 nF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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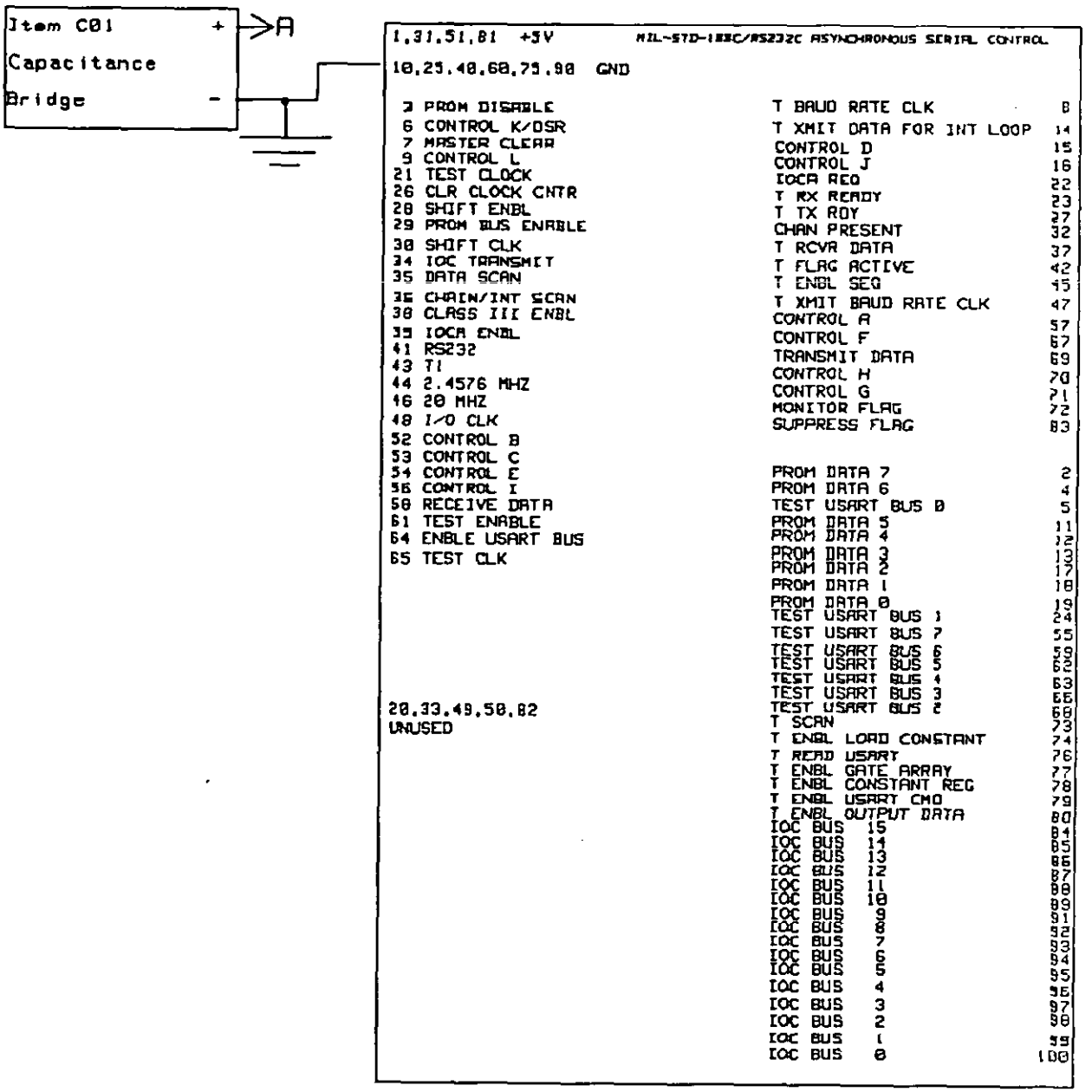


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	500	1500	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.25 A	NA	N
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	N N
Power supply current measurement	NA	± 50 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (8000 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (8000 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.25 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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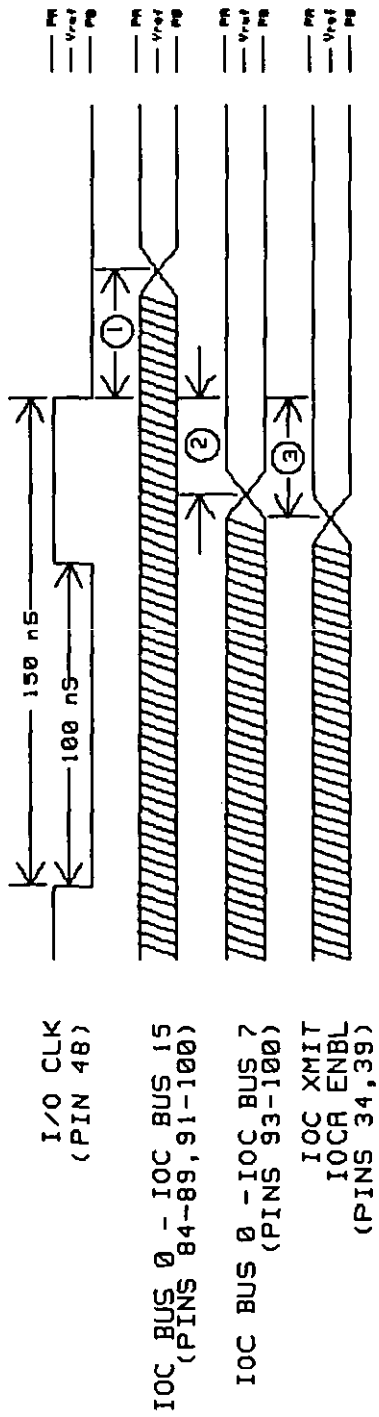


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) I/O CLK to IOC BUS 0 - IOC BUS 15. Hold time: 47 nS minimum.

2. (2) IOC BUS 0 - IOC BUS 7 to I/O CLK. Setup time: 24 nS minimum.

3. (3) IOCA ENBL to I/O CLK, IOC XMIT to I/O CLK. Setup time: 30 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (8000 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (8000 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (8000 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (8000 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.25 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

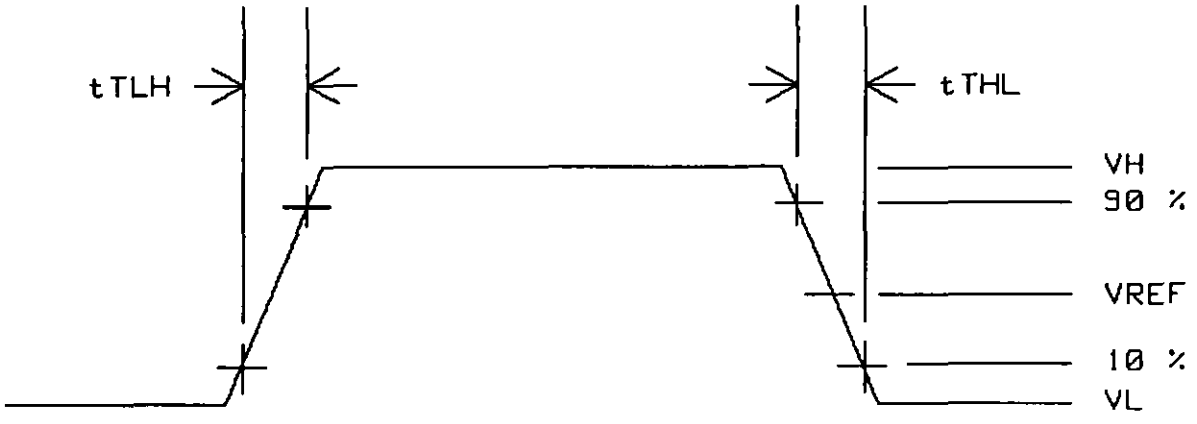
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	th	I/O CLK to IOC BUS 15	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 14	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 13	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 12	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 11	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 10	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 9	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 8	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 7	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 6	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 5	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 4	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 3	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 2	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 1	47	47	47	-	-	-	nS	12
1	th	I/O CLK to IOC BUS 0	47	47	47	-	-	-	nS	12
2	tsu	IOC BUS 7 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 6 to I/O CLK	24	24	24	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tsu	IOC BUS 5 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 4 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 3 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 2 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 1 to I/O CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 0 to I/O CLK	24	24	24	-	-	-	nS	12
3	tsu	IOCA ENBL to I/O CLK	30	30	30	-	-	-	nS	12
3	tsu	IOC XMIT to I/O CLK	30	30	30	-	-	-	nS	12
4	tpd	20 MHZ to PROM DATA 7	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 6	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 5	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 4	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 3	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 2	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 1	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to PROM DATA 0	-	-	-	250	250	250	nS	12
5	tpd	20 MHZ to T SCAN	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL CNST	-	-	-	128	128	128	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	20 MHZ to T RD USART	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL ARRAY	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL REG	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL USART	-	-	-	128	128	128	nS	12
5	tpd	20 MHZ to T ENBL DATA	-	-	-	128	128	128	nS	12
6	tpd	20 MHZ to TEST USART 0	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 1	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 7	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 6	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 5	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 4	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 3	-	-	-	405	405	405	nS	12
6	tpd	20 MHZ to TEST USART 2	-	-	-	405	405	405	nS	12
7	tpd	DATA SCAN to IOCA REQ	-	-	-	96	96	96	nS	12
8	tpd	TEST ENBL to BAUD RATE	-	-	-	25	25	25	nS	12
8	tpd	TEST ENBL to T ENBL SEQ	-	-	-	25	25	25	nS	12
8	tpd	TEST ENBL to T XMIT BAUD	-	-	-	25	25	25	nS	12

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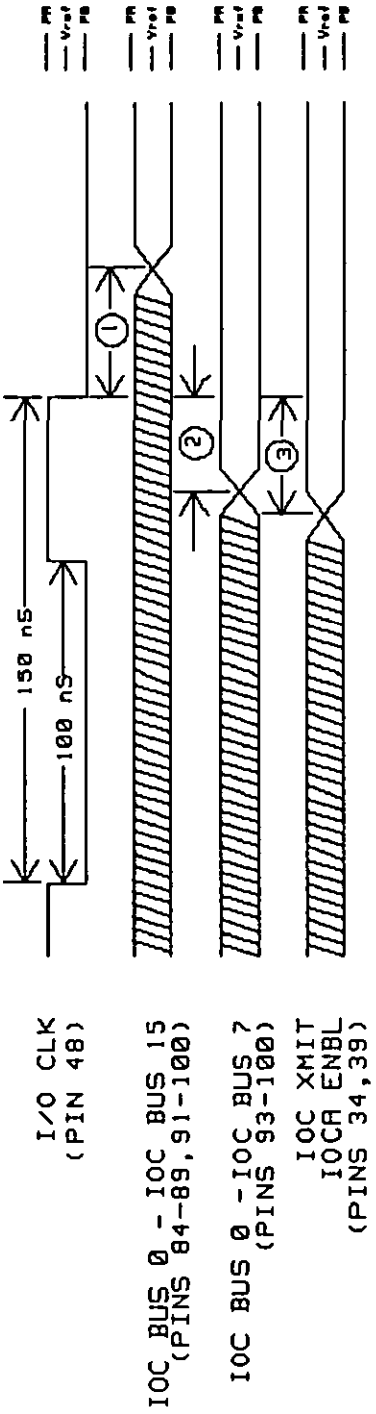


FIGURE 12a. Dynamic waveforms for table XVII.

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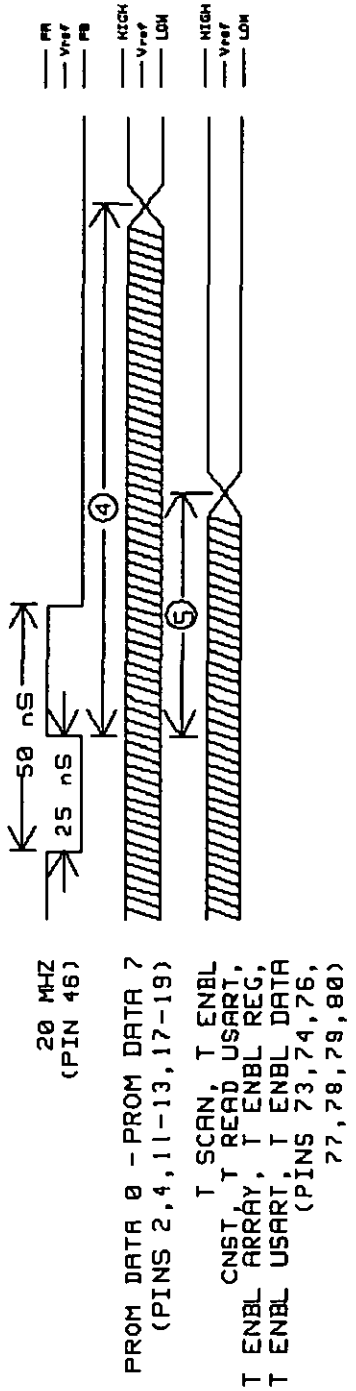


FIGURE 12b. Dynamic waveforms for table XVII.

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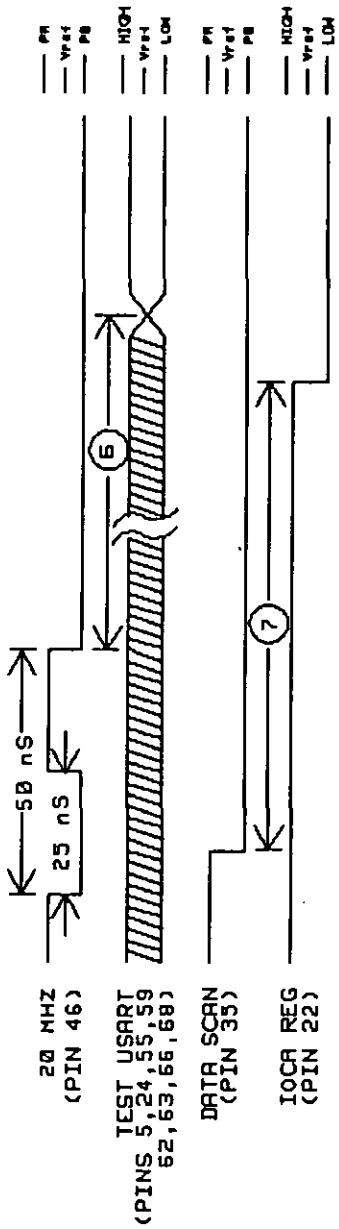


FIGURE 12c. Dynamic waveforms for table XVII.

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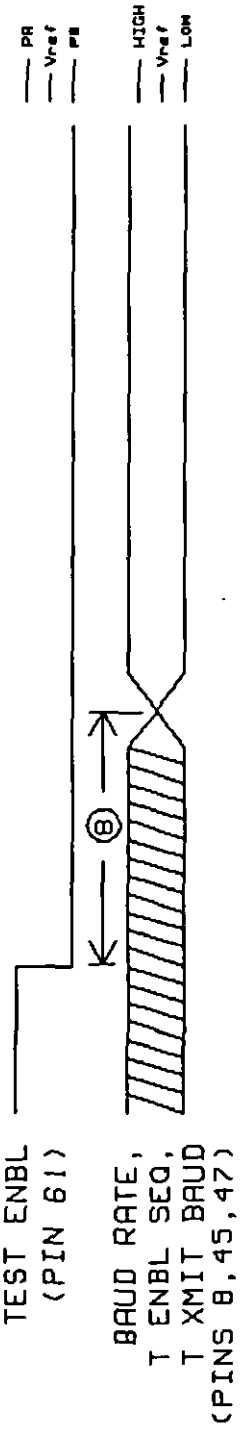


FIGURE 12d. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
Inputs						
R1	PROM DISABLE	3	6	175	pF	1-4
R2	CTRL K/DSR	6	2	75	pF	1-4
R3	MSTR CLR	7	2	100	pF	1-4
R4	CTRL L	9	2	100	pF	1-4
R5	TEST CLK	21	2	75	pF	1-4
R6	CLR CLK CNTR	26	6	100	pF	1-4
R7	SHIFT ENBL	28	4	75	pF	1-4
R8	PROM ENBL	29	2	100	pF	1-4
R9	SHIFT CLK	30	4	75	pF	1-4
R10	IOC XMIT	34	2	75	pF	1-4
R11	DATA SCAN	35				
R12	CHN/INT SCAN	36				
R13	CL III ENBL	38				
R14	IOCA ENBL	39	2	75	pF	1-4
R15	RS232	41	2	125	pF	1-4
R16	T1	43	2	75	pF	1-4
R17	2.4576 MHZ	44				
R18	20 MHZ	46	2	75	pF	1-4
R19	I/O CLK	48	2	125	pF	1-4
R20	CTRL B	52	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CTRL C	53	2	75	pF	1-4
R22	CTRL E	54				
R23	CTRL I	56				
R24	RCV DATA	58	2	75	pF	1-4
R25	TEST ENBL	61	8	200	pF	1-4
R26	ENBL USART	64	2	75	PF	1-4
R27	TEST CLK	65	2	75	pF	1-4
Bidirectionals						
R28	PROM DATA 7	2	2	75	pF	1-4
R29	PROM DATA 6	4	2	75	pF	1-4
R30	TEST USART 0	5	2	100	pF	1-4
R31	PROM DATA 5	11	2	75	pF	1-4
R32	PROM DATA 4	12				
R33	PROM DATA 3	13				
R34	PROM DATA 2	17	2	75	pF	1-4
R35	PROM DATA 1	18	2	100	pF	1-4
R36	PROM DATA 0	19				
R37	TEST USART 1	24	2	100	pF	1-4
R38	TEST USART 7	55	2	75	pF	1-4
R39	TEST USART 6	59				
R40	TEST USART 5	62	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	TEST USART 4	63	2	75	pF	1-4
R42	TEST USART 3	66				
R43	TEST USART 2	68				
R44	T SCAN	73				
R45	T ENBL CNST	74				
R46	T RD USART	76				
R47	T ENBL ARRAY	77				
R48	T ENBL REG	78				
R49	T ENBL USART	79				
R50	T ENBL DATA	80	2	75	pF	1-4
R51	IOC BUS 15	84	3	75	pF	1-4
R52	IOC BUS 14	85				
R53	IOC BUS 13	86				
R54	IOC BUS 12	87				
R55	IOC BUS 11	88				
R56	IOC BUS 10	89				
R57	IOC BUS 9	91				
R58	IOC BUS 8	92				
R59	IOC BUS 7	93				
R60	IOC BUS 6	94	3	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R61	IOC BUS 5	95	3	75	pF	1-4
R62	IOC BUS 4	96				
R63	IOC BUS 3	97				
R64	IOC BUS 2	98				
R65	IOC BUS 1	99				
R66	IOC BUS 0	100	3	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R66
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R66
Capacitance measurement	NA	± 8.45 pF	R1
	NA	± 3.65 pF	R2
	NA	± 4.9 pF	R3,R4
	NA	± 3.65 pF	R5
	NA	± 4.7 pF	R6
	NA	± 3.55 pF	R7
	NA	± 4.9 pF	R8
	NA	± 3.55 pF	R9
	NA	± 3.65 pF	R10-R14
	NA	± 6.15 pF	R15
	NA	± 3.65 pF	R16-R18
	NA	± 6.15 pF	R19
	NA	± 3.65 pF	R20-R24
	NA	± 9.6 pF	R25
	NA	± 3.65 pF	R26-R29
	NA	± 4.9 pF	R30
	NA	± 3.65 pF	R31-R34
	NA	± 4.9 pF	R35-R37
	NA	± 3.65 pF	R38-R50
	NA	± 3.6 pF	R51-R66

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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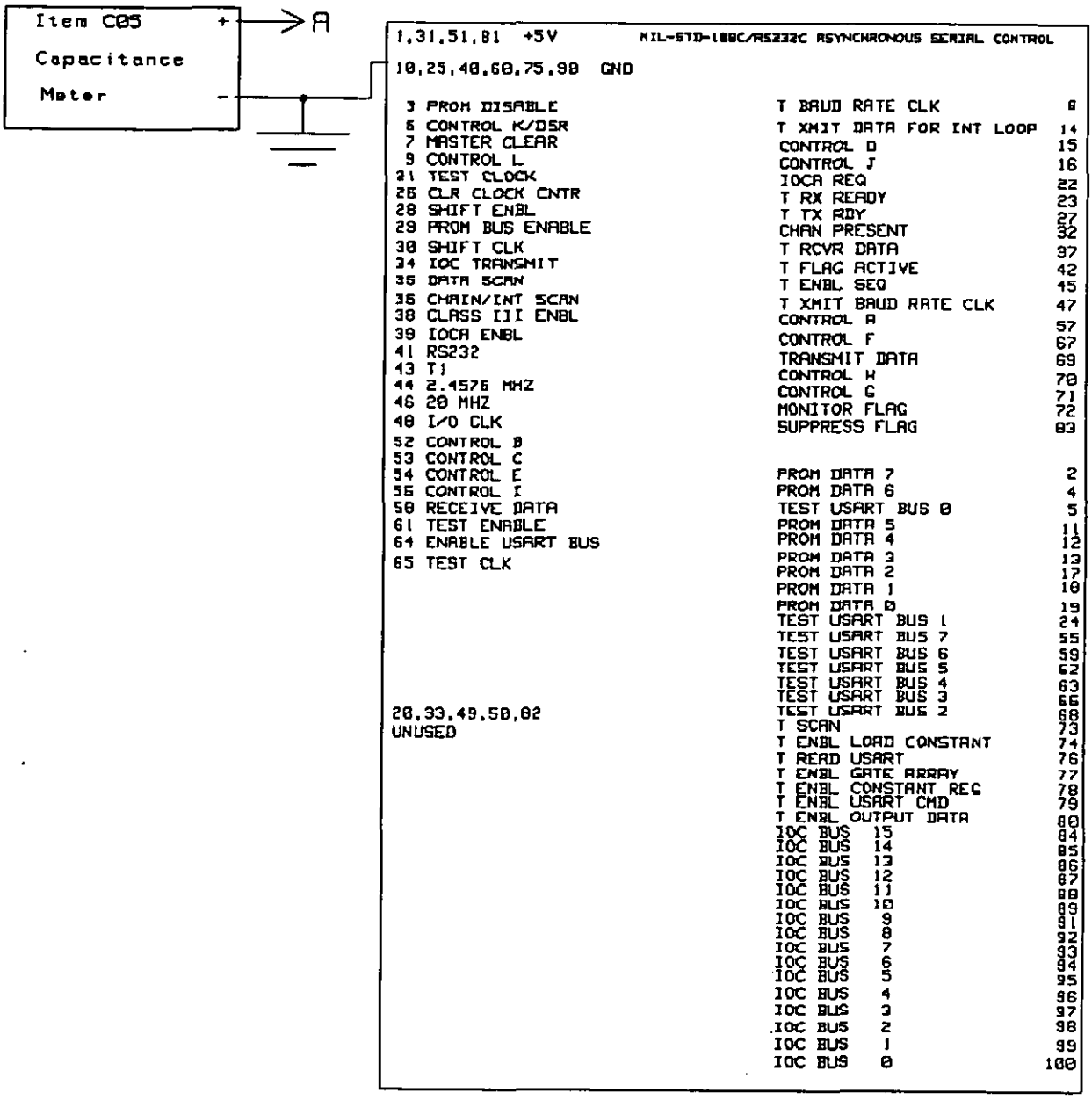


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	2.25 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', connected to Vcc at the point of shorting.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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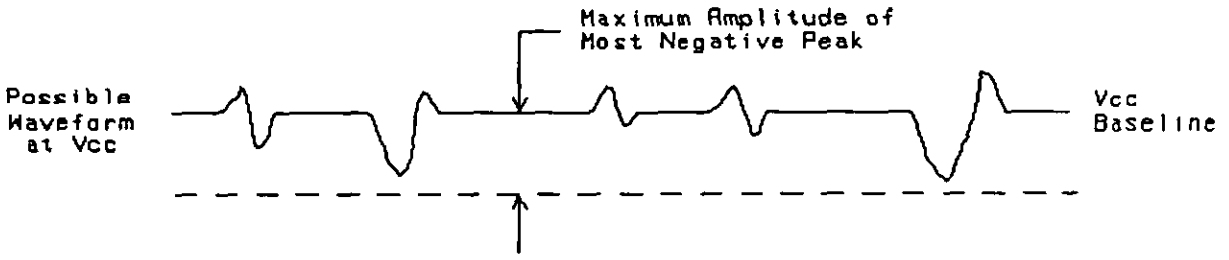
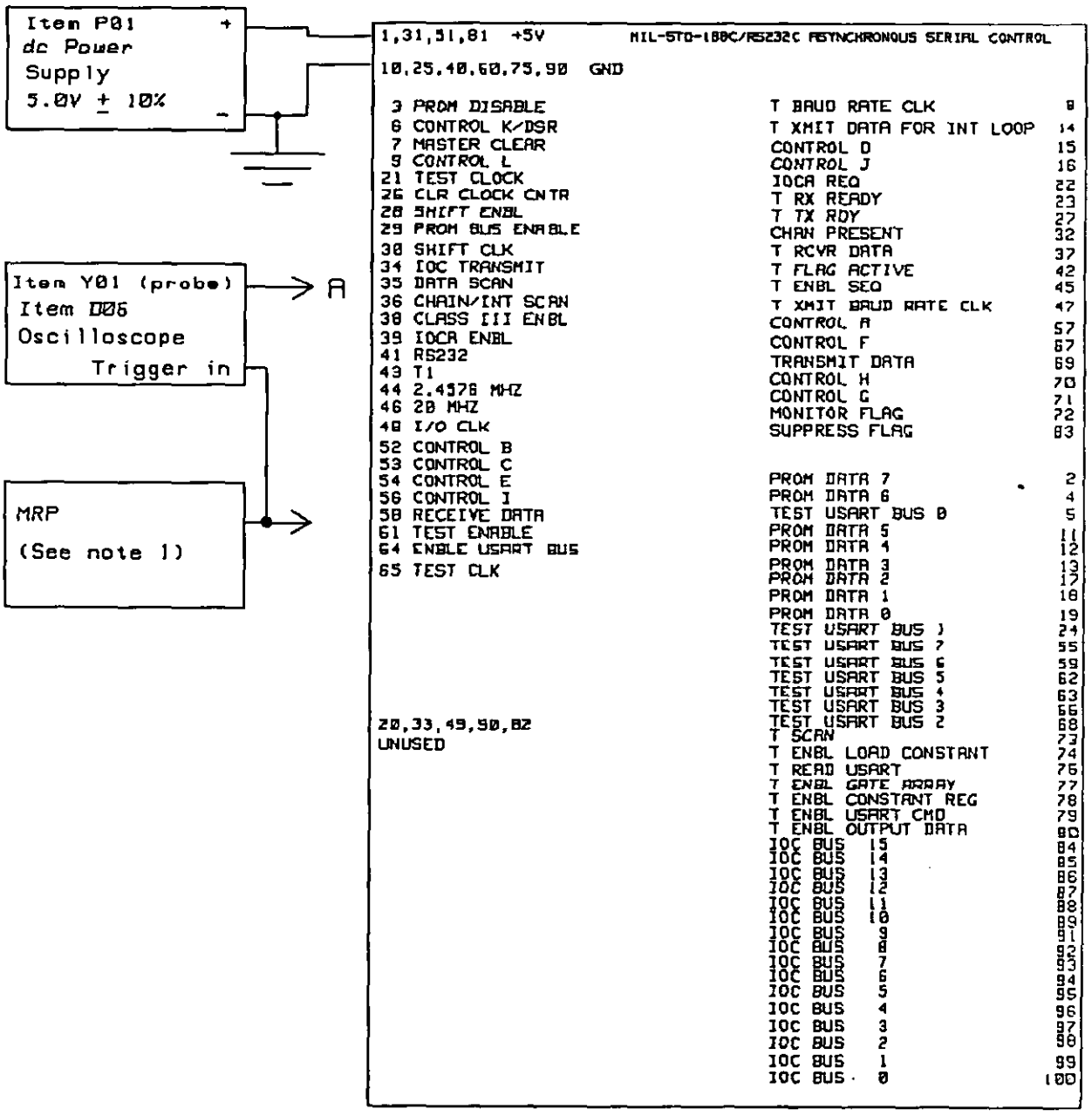


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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